38) Construct a truth table for each of these compound propositions.

a) $(p \vee q) \vee r$
b) $(p \vee q) \wedge r$
c) $(p \wedge q) \vee r$
d) $(p \wedge q) \wedge r$
e) $(p \vee q) \wedge \neg r$
f) $(p \wedge q) \vee \neg r$

p	q	r	$(p \vee q)$	$(p \vee q) \vee r$	$(p \vee q) \wedge r$	$(p \wedge q)$
T	T	T	T	T	T	T
T	T	F	T	T	F	T
T	F	T	T	T	T	F
T	F	F	T	T	F	F
F	T	T	T	T	T	F
F	T	F	T	T	F	F
F	F	T	F	T	T	F
F	F	F	F	F	F	F

$(p \wedge q) \vee r$	$(p \wedge q) \wedge r$	$\neg r$	$(p \vee q) \wedge \neg r$	$(p \wedge q) \vee \neg r$
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
T	F	T	T	T
T	F	F	F	F
T	F	T	T	T
F	F	F	F	F
F	F	T	F	T

39) Construct a truth table for each of these compound propositions

a) $p \rightarrow (\neg q \vee r)$
b) $\neg p \rightarrow (q \rightarrow r)$

p	q	r	$\neg q$	$(\neg q \vee r)$	$p \rightarrow (\neg q \vee r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	T	T

p	q	r	$\neg p$	$q \rightarrow r$	$\neg p \rightarrow (q \rightarrow r)$
T	T	T	F	T	T
T	T	F	F	F	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

→ F F F T T T